Mandeep Khatri

Huntsville, AL | (202) 423-9113

mandeep.khatri@bulldogs.aamu.edu | mandyy.me | github.com/Mandeep-Khatri | linkedin.com/in/mandeep-khatri

EDUCATION

Alabama A&M University, *Huntsville, AL* — ***Expected Graduation: May 2028***

Bachelor of Science in Computer Science | Minor: Finance — *GPA: 3.85/4.0*

Awards & Honors: Dean's List, AAMU Merit Full Scholarship, EICOP 2025 Finalist

Relevant Coursework: Data Structures & Algorithms, Intro to programming I and II (Python), Computer Architecture, Software Engineering, Operating Systems, Discrete Mathematics, Linear Algebra

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, JavaScript, TypeScript, SQL, MATLAB, Go

Frameworks & Libraries: React, React Native, NestJS, Flask, pandas, NumPy, Polars, Streamlit, LangChain, FAISS

Cloud & Tools: AWS (EC2, Lambda), GCP (Vertex AI), Docker, PostgreSQL, Prisma, Git, GitLab CI, CUDA, Linux

WORK EXPERIENCE

Research Intern - International Space Weather Camp, *German Aerospace Center (DLR)* — ***June 2026 – Present***

- Built a **Python** and **Polars pipeline** to clean and align 20M+ rows of Wind and STEREO spacecraft data, yielding a dataset of 752 quality-filtered CME events for model development.
- Engineered 56 statistical features and evaluated **Linear Regression, Random Forest,** and Gradient Boosting across 20 seeds; the best model reached 0.93 Pearson correlation for maximum magnetic field strength (Bt).

Teaching Assistant, *Alabama A&M University, Huntsville, AL* — **October 2025 – Present**

- Built a **pytest autograder** for CS104 (Python) and CS109 (C++), adopted by the TA team across 35-student lab sections; cut per-assignment grading from 2.5h to 25min and rewrote 4 lab handouts after student feedback.
- Started a weekly LeetCode workshop for freshmen prepping for internships; ran a 12-week DSA track (arrays, hash maps, graphs, DP) on GitHub Classroom with mock-interview sessions, growing to 15+ regulars.

Research Engineer - Autonomous System Lab, *Alabama A&M University, Huntsville, AL* — ***January 2025 – Present***

- Led a rewrite of the drone perception pipeline from **MATLAB** to **C++/CUDA**; optimized **parallel GPU processing** and memory access, reducing per-frame latency from **84 ms** to **27 ms** on Jetson Orin while sustaining 30 FPS.
- Developed Python automation scripts & **Gazebo** regression-testing framework covering 14 scenarios; integrated tests into GitLab CI to automate routine perception testing & catch a frame-skip bug before flight testing.
- Proposed and built a Kalman-filter sensor-fusion module in **C++** that smooths IMU + camera streams in the perception stack, cutting simulated trajectory jitter 38%.

Software Engineering Intern, *Verisk, Remote* — ***August 2023 – December 2023***

- Added new sensor types in the team's IoT telemetry service (Python, MQTT, AWS EC2); shipped a JavaScript filter on the Chart.js dashboard, a Flask CSV endpoint, 47 PyTest cases, & helped Dockerize the service.
- Investigated slow REST API responses under JMeter load testing (~300 RPS) and worked with my mentor to identify a Postgres connection pool issue; after the fix, p95 API latency dropped from ~1.4s to ~820ms.

PROJECTS

Route Intelligence - *Food Rescue Platform* — ***JPMorgan Chase Hackathon, 2026***

- Processed 150K+ rescue records (6.67M lbs of food) using Python and Pandas across 4 cities; a ZIP-3 batching pass flagged 845 same-day pickup pairs, unlocking $106K in potential logistics savings.
- Built a 4-tier risk score (recency, claim rate, pickup size) on a Folium + Streamlit map; 35% of rescues fell into Watch/Elevated/High.

Internably – *Student Networking Platform* — ***Present***

- Built a full-stack mobile application for college internship networking using React Native, TypeScript, NestJS, PostgreSQL, and Prisma; designed authentication with .edu verification, JWT refresh-token rotation, and real-time messaging through Socket.IO.
- Set up an npm workspaces monorepo with three shared packages (UI tokens, TS types, ESLint config) for the mobile and planned web client; TanStack Query for caching, Zustand for state.

LLM Course Q&A Chatbot — ***October 2025***

- Built a **machine learning and information retrieval** RAG pipeline that ingested 100+ pages of course PDFs using PyMuPDF, Gemini embeddings, Vertex AI, and FAISS, achieving 85% manual Q&A accuracy.
- Optimized FAISS dense retrieval and prompt engineering templates, reducing average query response latency by 40% compared to a keyword search baseline, improving real-time usability for students.

ACTIVITIES

- **Break Through Tech AI Fellow, Cornell Tech** - *Applied AI/ML program* — **May 2026 - Present**
- **NRF Foundation Ray Greenly Scholar** - *Competitive National Selection* — **January 2026**
- **Google Cloud Career Jumpstart** - *HBCU Fellow* — **May 2025 – August 2025**
- **Propel2Excel Fellow** - *selective mentorship cohort* — **September 2025 – Present**
- **TracerFire Cybersecurity Competition** - *Competitor, Sandia National Laboratories* — **September 2024**